Block practice · Reading code: constructs, named items, errors and test data

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Syntax, logic and runtime errors

Spot constructs, operators, keywords and comments

Find named items in a program

Normal, boundary and erroneous test data

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Practice workbook · Syntax, logic and runtime errors

Defining logic, syntax and runtime errors, classifying described faults, naming the error from symptoms and giving an example.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T2-08

1. Question 1 · part b.i · 3 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b.i only.

1 Programmers use different programming constructs to create working code.

(a) Identify the keyword used to create a selection statement.

(1)

- ☒ **A** for
- ☒ **B** if
- ☒ **C** new
- ☒ **D** while

(b) Computer code can contain errors.

(i) Complete the table by adding a tick (✓) to match each error description to a type of error.

(3)

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.			
The program does not translate.			
The program crashes during execution.			

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1. Question 1 · part b.i · 3 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b.i only.

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DO NOT WRITE IN THIS AREA

(ii) A program counts the number of green and red balls stored in an array.

The program outputs the number of green balls, the number of red balls and the total number of balls.

Figure 1 shows the expected output from the program.

```
Green:  25 balls
Red:  15 balls
Total:  40 balls
```

Figure 1

Open **Q01bii** in the code editor.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01biiFINISHED** with the correct file extension for the programming language.

(3)

(c) Programs use constants, variables and data structures.

(i) Give **one** reason for using a constant.

(1)

(ii) Describe **one** difference between a variable and an array.

(2)

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1. Question 1 · part b.i · 3 marks · Working and answer

Working and answer

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2. Question 1 · part b · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part b only.

1 Computer programs use several different programming constructs.

(a) Identify the **most appropriate** data type for the value 4.5

(1)

- ☒ **A** Boolean
- ☒ **B** Character
- ☒ **C** Real
- ☒ **D** String

(b) Give a definition of a logic error.

(1)

(c) Open **Q01c** in the code editor.

The program should print out a counter and the counter with 7 added to it.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) Give **one** reason why computer programmers should use comments in their programs.

(1)

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2. Question 1 · part b · 1 mark · Working and answer

Working and answer

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3. Question 1 · part b · 3 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

1 Programmers write code to solve problems.

(a) Give **one** technique that programmers use to make their code easy to read and understand.

(1)

.....

(b) Programs may contain logic, syntax or runtime errors.

A program has three errors.

These may be logic, syntax or runtime errors.

Complete the table by adding a tick (✓) for **each** error description to show whether the error is a logic, syntax or runtime error.

(3)

Error description	Logic	Syntax	Runtime
Divide by 0			
Use x instead of * to multiply			
Subtract 10 from 2 instead of 2 from 10			

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3. Question 1 · part b · 3 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

(c) **Figure 1** shows a programming construct in a flowchart.

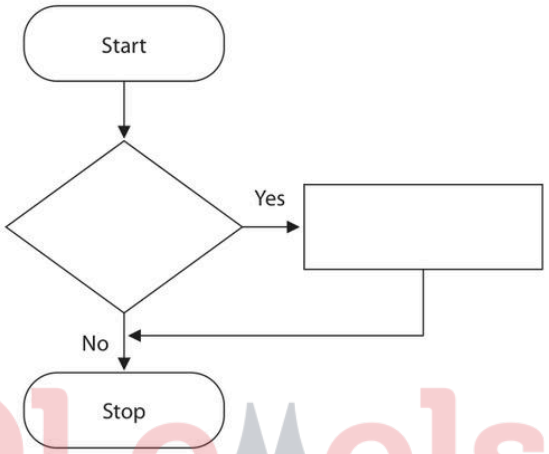


Figure 1

Identify the name of the programming construct.

(1)

- ☒ **A** Array
- ☒ **B** Iteration
- ☒ **C** Operator
- ☒ **D** Selection

(d) Open **Q01d** in the code editor.

The program inputs two numbers, compares them and outputs a suitable message.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01dFINISHED** with the correct file extension for the programming language.

(3)

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3. Question 1 · part b · 3 marks · Working and answer

Working and answer

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4. Question 1 · part e · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part e only.

1 Programmers write code to solve problems.

(a) Identify the description of a variable in a computer program.

(1)

- ☒ **A** A value that cannot be used more than once
- ☒ **B** A value that must be input
- ☒ **C** A value that is always used in a calculation
- ☒ **D** A value that can change

(b) Identify the technique that improves the readability of code.

(1)

- ☒ **A** Using indents on every line
- ☒ **B** Using descriptive names for variables
- ☒ **C** Using the correct operators
- ☒ **D** Using suitable data structures

(c) Complete the table by giving an example value for **each** data type.

The first row has been completed for you.

(2)

Data type	Example value
integer	12
char	
real	

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4. Question 1 · part e · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part e only.

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(d) Describe **one** difference between the data used in boundary testing and the data used in erroneous testing.

(2)

(e) A program should output the value 2

However, there is an error in the code and the actual output is 4

Name this type of error.

(1)

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(f) Open **Q01f** in the code editor.

The program should allow the input of the length of the side of a square and output the area of the square.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01fFINISHED** with the correct file extension for the programming language.

(3)

(g) A program is needed that will accept the input of a letter and compare it with a stored letter.

It will check whether the letter comes earlier in the alphabet, later in the alphabet or is the same letter as the stored letter.

It will output the letter and the stored letter with an appropriate message.

Open **Q01g** in the code editor.

Amend the code to complete the if statement used to produce the output.

You must use the structure given in **Q01g** to complete the program.

Do not add any further functionality.

Save your code as **Q01gFINISHED** with the correct file extension for the programming language.

(4)

(Total for Question 1 = 14 marks)

4. Question 1 · part e · 1 mark · Working and answer

Working and answer

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5. Question 2 · part b.i · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part b.i only.

2 Programmers design, correct and test programs.

(a) Programmers design programs.

(i) Identify the term for a step-by-step description to complete a task.

(1)

☒ **A** Algorithm

☒ **B** Computational thinking

☒ **C** Decomposition

☒ **D** Pseudocode

(ii) A constant is a memory location whose value does not change during program execution.

Give the name for a memory location whose value can change during program execution.

(1)

.....

.....

(b) Programs can have runtime errors.

(i) State what is meant by the term **runtime error**.

(1)

.....

.....

(ii) Give **one** example of a runtime error.

(1)

.....

.....

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5. Question 2 · part b.i · 1 mark · Working and answer

Working and answer

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6. Question 2 · part b.ii · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part b.ii only.

2 Programmers design, correct and test programs.

(a) Programmers design programs.

(i) Identify the term for a step-by-step description to complete a task.

(1)

- ☒ **A** Algorithm
- ☒ **B** Computational thinking
- ☒ **C** Decomposition
- ☒ **D** Pseudocode

(ii) A constant is a memory location whose value does not change during program execution.

Give the name for a memory location whose value can change during program execution.

(1)

(b) Programs can have runtime errors.

(i) State what is meant by the term **runtime error**.

(1)

(ii) Give **one** example of a runtime error.

(1)

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6. Question 2 · part b.ii · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part b.ii only.

(c) Open **Q02c** in the code editor.

The program generates a random number for the width of a rectangle.

The length of the rectangle is always 4

The program should calculate the perimeter of the rectangle.

The program should display the width, the length and the perimeter of the rectangle.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q02cFINISHED** with the correct file extension for the programming language.

(3)

(d) **Figure 1** shows part of an online form displayed in a browser.

Memorable word		(2-8 letters)
Date	31-01-2019	

Figure 1

Complete the table to show **two** examples of erroneous, **one** example of normal and **one** example of boundary test data for the **memorable word** field.

Each example must be different.

(4)

Erroneous	
Erroneous	
Normal	
Boundary	

(Total for Question 2 = 11 marks)

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6. Question 2 · part b.ii · 1 mark · Working and answer

Working and answer

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Practice workbook · Spot constructs, operators, keywords and comments

Reading opened Python code, printed pseudocode or a lettered list of algorithm steps and pointing to an element: a relational or logical operator, the selection keyword, comment text, the line where selection or iteration starts, a line that defines a subprogram, initialises a variable or takes input, including marking such a line with a comment in the editor; also recognising arithmetic, relational and logic operators and why brackets change the order of operations.

23 question parts: 1 worked-example retry and 22 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T2-13

1. Question 1 · part c.iv · 1 mark

2024 JUN · Paper 02

Worked-example retry · Attempt part c.iv only.

1 Programmers write programs to solve problems.

(a) Programmers use many different techniques and tools to solve problems.

Identify what is meant by the term **abstraction**.

(1)

- ☒ **A** Breaking a problem down into smaller more manageable parts
- ☒ **B** Deciding if a program is efficient in terms of execution time
- ☒ **C** Drawing flowcharts with symbols used in the computer industry
- ☒ **D** Removing unnecessary detail to highlight the important points

(b) Open **Q01b** in the code editor.

A user enters a name and an age when the program executes.

The program should display a welcome message when the user is less than 30 years of age.

Amend the code to complete the program.

Save your amended code as **Q01bFINISHED** with the correct file extension for the programming language.

(4)

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1. Question 1 · part c.iv · 1 mark

2024 JUN · Paper 02

Worked-example retry · Attempt part c.iv only.

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(c) Open **Q01c** in the code editor.

Use the code to answer these questions.

(i) Give the text of a comment used in the code.

(1)

(ii) Give the text of a line that creates and initialises a variable.

(1)

(iii) Give the keyword that starts the selection used in the code.

(1)

(iv) Give the logical operator used in the code.

(1)

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(d) Arrays and records are data structures.

Identify the data that **must be** stored in a record rather than an array.

(1)

- ☒ **A** "King Lear", "Macbeth", "Romeo and Juliet", "The Tempest"
- ☒ **B** "King Lear", 1983, 78.32, 99
- ☒ **C** 1978, 1985, 1990, 2001
- ☒ **D** 13.99, 12.75, 10.58, 11.43

(e) Complete the table to show the data type for **each** item.

(2)

Item	Data type
45.82	
True	

(Total for Question 1 = 12 marks)

1. Question 1 · part c.iv · 1 mark · Working and answer

Working and answer

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2. Question 2 · part b.ii · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part b.ii only.

2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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2. Question 2 · part b.ii · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part b.ii only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

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2. Question 2 · part b.ii · 1 mark · Working and answer

Working and answer

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3. Question 2 · part c.i · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.i only.

2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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3. Question 2 · part c.i · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.i only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

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3. Question 2 · part c.i · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.i only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

(c) Anna has written a program that manipulates text.

Open **Q02c** in the code editor.

Use the code to answer these questions.

(i) Identify a line number where the code includes a relational operator.

(1)

(ii) Identify a line number where a variable is set to a numeric value entered by a user.

(1)

(iii) Identify the name of a variable passed from the main program to the subprogram.

(1)

(iv) Identify the name of a variable that receives the value calculated by the subprogram.

(1)

(v) State the value returned by the subprogram when the input values are **educational**, **3**, **6**

(2)

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3. Question 2 · part c.i · 1 mark · Working and answer

Working and answer

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4. Question 2 · part c.ii · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.ii only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

4. Question 2 · part c.ii · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.ii only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

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4. Question 2 · part c.ii · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.ii only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

(c) Anna has written a program that manipulates text.

Open **Q02c** in the code editor.

Use the code to answer these questions.

(i) Identify a line number where the code includes a relational operator.

(1)

(ii) Identify a line number where a variable is set to a numeric value entered by a user.

(1)

(iii) Identify the name of a variable passed from the main program to the subprogram.

(1)

(iv) Identify the name of a variable that receives the value calculated by the subprogram.

(1)

(v) State the value returned by the subprogram when the input values are **educational**, **3**, **6**

(2)

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4. Question 2 · part c.ii · 1 mark · Working and answer

Working and answer

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5. Question 1 · part a · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part a only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is **not** an arithmetic operator.

(1)

- ☒ **A** add
- ☒ **B** divide
- ☒ **C** equals
- ☒ **D** modulus

(b) State the values used by the Boolean data type.

(1)

.....

.....

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DO NOT WRITE IN THIS AREA

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5. Question 1 · part a · 1 mark · Working and answer

Working and answer

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6. Question 1 · part e.i · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part e.i only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is **not** an arithmetic operator.

(1)

- ☒ **A** add
- ☒ **B** divide
- ☒ **C** equals
- ☒ **D** modulus

(b) State the values used by the Boolean data type.

(1)

.....

.....

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DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

6. Question 1 · part e.i · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part e.i only.

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DO NOT WRITE IN THIS AREA

(c) Open **Q01c** in the code editor.

The program should multiply two integers.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) State what is meant by the term 'variable'.

(1)

(e) Ayesha has written a program that calculates the sum of a set of numbers.

Open **Q01e** in the code editor.

Answer these questions about the code.

(i) Identify the line number where the code includes a logic operator.

(1)

(ii) Identify the line number where a subprogram definition starts.

(1)

(iii) Identify the name of a numeric variable.

(1)

(iv) Identify the name of a string variable.

(1)

(Total for Question 1 = 10 marks)

6. Question 1 · part e.i · 1 mark · Working and answer

Working and answer

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7. Question 1 · part e.ii · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part e.ii only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is **not** an arithmetic operator.

(1)

- ☒ **A** add
- ☒ **B** divide
- ☒ **C** equals
- ☒ **D** modulus

(b) State the values used by the Boolean data type.

(1)

.....

.....

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

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7. Question 1 · part e.ii · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part e.ii only.

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DO NOT WRITE IN THIS AREA

(c) Open **Q01c** in the code editor.

The program should multiply two integers.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) State what is meant by the term 'variable'.

(1)

(e) Ayesha has written a program that calculates the sum of a set of numbers.

Open **Q01e** in the code editor.

Answer these questions about the code.

(i) Identify the line number where the code includes a logic operator.

(1)

(ii) Identify the line number where a subprogram definition starts.

(1)

(iii) Identify the name of a numeric variable.

(1)

(iv) Identify the name of a string variable.

(1)

(Total for Question 1 = 10 marks)

7. Question 1 · part e.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 1 · part a · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part a only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is a logic operator.

(1)

- ☒ **A** modulus
- ☒ **B** OR
- ☒ **C** >=
- ☒ **D** =

(b) Describe **one** difference between a one-dimensional array and a two-dimensional array.

(2)

(c) Open **Q01c** in the code editor.

The program should calculate how many £20 are in £113 and how much is left over.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

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8. Question 1 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 1 · part d.i · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.i only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is a logic operator.

(1)

- ☒ **A** modulus
- ☒ **B** OR
- ☒ **C** >=
- ☒ **D** =

(b) Describe **one** difference between a one-dimensional array and a two-dimensional array.

(2)

(c) Open **Q01c** in the code editor.

The program should calculate how many £20 are in £113 and how much is left over.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

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9. Question 1 · part d.i · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.i only.

DO NOT WRITE IN THIS AREA

(d) Raoul has written a program that counts the number of vowels in a sentence.

Open **Q01d** in the code editor.

Amend the code by adding a suitable comment (you may need to add more than one comment to a line):

(i) at the end of the line where there is a **relational operator**

(1)

(ii) at the end of a line where **iteration** starts

(1)

(iii) at the end of the line where **selection** starts

(1)

(iv) at the end of a line where a **data structure** is initialised.

(1)

Save your amended code as **Q01dFINISHED** with the correct file extension for the programming language.

(Total for Question 1 = 10 marks)

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Working and answer

10. Question 1 · part d.ii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.ii only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is a logic operator.

(1)

- ☒ **A** modulus
- ☒ **B** OR
- ☒ **C** >=
- ☒ **D** =

(b) Describe **one** difference between a one-dimensional array and a two-dimensional array.

(2)

(c) Open **Q01c** in the code editor.

The program should calculate how many £20 are in £113 and how much is left over.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

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10. Question 1 · part d.ii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.ii only.

DO NOT WRITE IN THIS AREA

(d) Raoul has written a program that counts the number of vowels in a sentence.

Open **Q01d** in the code editor.

Amend the code by adding a suitable comment (you may need to add more than one comment to a line):

(i) at the end of the line where there is a **relational operator**

(1)

(ii) at the end of a line where **iteration** starts

(1)

(iii) at the end of the line where **selection** starts

(1)

(iv) at the end of a line where a **data structure** is initialised.

(1)

Save your amended code as **Q01dFINISHED** with the correct file extension for the programming language.

(Total for Question 1 = 10 marks)

OLevels.com SCHOOL

Working and answer

11. Question 1 · part d.iii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.iii only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is a logic operator.

(1)

- ☒ **A** modulus
- ☒ **B** OR
- ☒ **C** >=
- ☒ **D** =

(b) Describe **one** difference between a one-dimensional array and a two-dimensional array.

(2)

(c) Open **Q01c** in the code editor.

The program should calculate how many £20 are in £113 and how much is left over.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

11. Question 1 · part d.iii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.iii only.

DO NOT WRITE IN THIS AREA

(d) Raoul has written a program that counts the number of vowels in a sentence.

Open **Q01d** in the code editor.

Amend the code by adding a suitable comment (you may need to add more than one comment to a line):

(i) at the end of the line where there is a **relational operator**

(1)

(ii) at the end of a line where **iteration** starts

(1)

(iii) at the end of the line where **selection** starts

(1)

(iv) at the end of a line where a **data structure** is initialised.

(1)

Save your amended code as **Q01dFINISHED** with the correct file extension for the programming language.

(Total for Question 1 = 10 marks)

OLevels.com SCHOOL

Working and answer

12. Question 1 · part d.iv · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.iv only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is a logic operator.

(1)

- ☒ **A** modulus
- ☒ **B** OR
- ☒ **C** >=
- ☒ **D** =

(b) Describe **one** difference between a one-dimensional array and a two-dimensional array.

(2)

(c) Open **Q01c** in the code editor.

The program should calculate how many £20 are in £113 and how much is left over.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

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12. Question 1 · part d.iv · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.iv only.

DO NOT WRITE IN THIS AREA

(d) Raoul has written a program that counts the number of vowels in a sentence.

Open **Q01d** in the code editor.

Amend the code by adding a suitable comment (you may need to add more than one comment to a line):

(i) at the end of the line where there is a **relational operator**

(1)

(ii) at the end of a line where **iteration** starts

(1)

(iii) at the end of the line where **selection** starts

(1)

(iv) at the end of a line where a **data structure** is initialised.

(1)

Save your amended code as **Q01dFINISHED** with the correct file extension for the programming language.

(Total for Question 1 = 10 marks)

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Working and answer

13. Question 4 · part b.i · 1 mark

2022 JUN · Paper 01

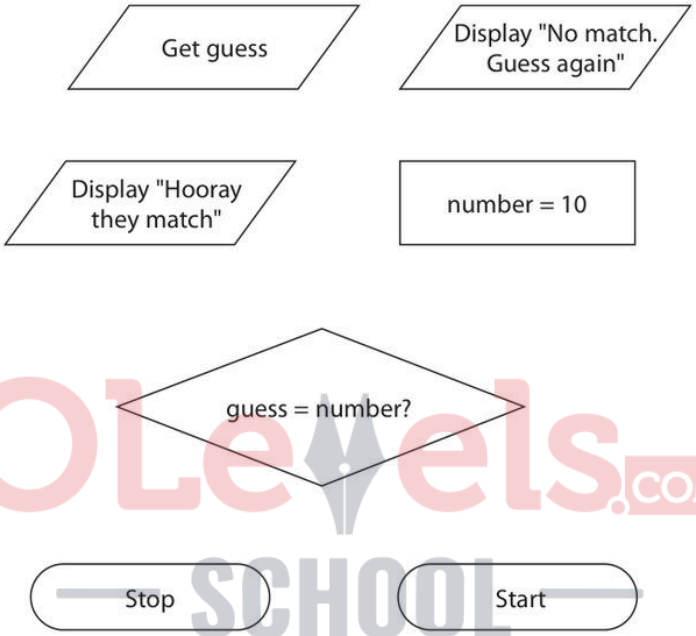
Further practice · Attempt part b.i only.

4 Reba likes writing programs.

(a) She is writing a guessing game.

She needs a flowchart to show the logic of the game.

(i) These are the components needed to draw the flowchart.



Draw the flowchart for the algorithm in the box on the next page.

Use each component once.

Do not add any additional components.

Use as many arrows and yes/no labels as you need.

(5)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 7 2 5 3 8 A 0 1 0 2 0

13. Question 4 · part b.i · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.i only.

DO NOT WRITE IN THIS AREA

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Draw your flowchart here.

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13. Question 4 · part b.i · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.i only.

(ii) Identify an alternative method for writing the algorithm.

(1)

- ☒ **A** Simulation
- ☒ **B** Cipher
- ☒ **C** Program code
- ☒ **D** Truth table

(b) Reba wants to develop a program that will convert a temperature in Fahrenheit to Celsius.

Here are four steps in the algorithm.

The steps are not in the correct order.

	Step
A	Change the temperature to Celsius
B	Get the temperature in Fahrenheit
C	Show the temperature in Celsius
D	Set the temperature to 0

(i) Give the letter of the step that initialises a variable.

(1)

.....

(ii) Give the letter of the step that inputs a value.

(1)

.....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 7 2 5 3 8 A 0 1 2 2 0

13. Question 4 · part b.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

14. Question 4 · part b.ii · 1 mark

2022 JUN · Paper 01

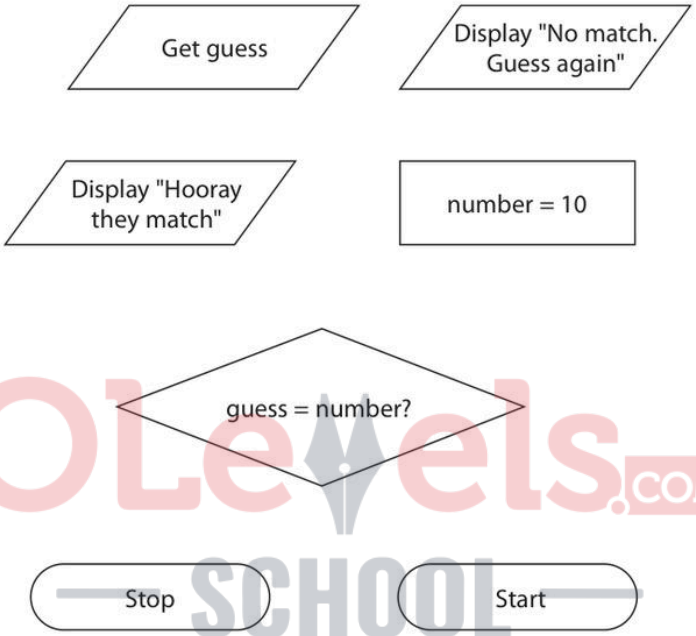
Further practice · Attempt part b.ii only.

4 Reba likes writing programs.

(a) She is writing a guessing game.

She needs a flowchart to show the logic of the game.

(i) These are the components needed to draw the flowchart.



Draw the flowchart for the algorithm in the box on the next page.

Use each component once.

Do not add any additional components.

Use as many arrows and yes/no labels as you need.

(5)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 7 2 5 3 8 A 0 1 0 2 0

14. Question 4 · part b.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

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Draw your flowchart here.

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14. Question 4 · part b.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.

(ii) Identify an alternative method for writing the algorithm.

(1)

- ☒ **A** Simulation
- ☒ **B** Cipher
- ☒ **C** Program code
- ☒ **D** Truth table

(b) Reba wants to develop a program that will convert a temperature in Fahrenheit to Celsius.

Here are four steps in the algorithm.

The steps are not in the correct order.

	Step
A	Change the temperature to Celsius
B	Get the temperature in Fahrenheit
C	Show the temperature in Celsius
D	Set the temperature to 0

(i) Give the letter of the step that initialises a variable.

(1)

.....

(ii) Give the letter of the step that inputs a value.

(1)

.....

DO NOT WRITE IN THIS AREA

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14. Question 4 · part b.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.

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QUESTION 4 (c) BEGINS ON THE NEXT PAGE.

14. Question 4 · part b.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.

(c) **Figure 2** shows the pseudocode for an early version of an algorithm that Reba has written for another game.

The algorithm:

- asks the user to input a colour or input –1 to end the game
- awards 1 point for red
- awards 8 points for orange
- generates the score for the game
- displays the results of the game.

```
SET Colour TO ""
SET Score TO 0
SET RedPoints TO 0
SET OrangePoints TO 0
SET NumOranges TO 0

WHILE Colour <> "-1" DO
   RECEIVE Colour FROM (STRING) KEYBOARD
   IF Colour = "red" THEN
      SET RedPoints TO RedPoints + 1
   ELSE
      IF Colour = "orange" THEN
         SET OrangePoints TO OrangePoints + 8
         SET NumOranges TO NumOranges + 1
      END IF
   END IF
END WHILE

SET Score TO RedPoints + OrangePoints

SEND ("Score: "& Score) TO DISPLAY
SEND ("Number of reds: "& RedPoints) TO DISPLAY
SEND ("Number of oranges: "& OrangePoints) TO DISPLAY
```

Figure 2

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14

P 7 2 5 3 8 A 0 1 4 2 0

14. Question 4 · part b.ii · 1 mark · Working and answer

Working and answer

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15. Question 2 · part a.i · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a.i only.

2 Raza is writing a program to tell the user whether a number they input is a prime number.

A prime number is a whole number, larger than one, that can only be divided by one and itself with no remainder.

This pseudocode contains the logic required to complete the program.

```
FUNCTION checkPrime(pNumber)
BEGIN FUNCTION
    IF pNumber = 1 THEN
        check = False
    ELSE
        check = True
        FOR count FROM 2 TO pNumber DO
            IF pNumber MOD count = 0 THEN
                check = False
            END IF
        END FOR
    END IF
RETURN check
END FUNCTION

SEND "Enter a number: " TO DISPLAY
RECEIVE number FROM (INTEGER) KEYBOARD
SET result TO checkPrime(number)

IF result = True THEN
    SEND (number & " is a prime number") TO DISPLAY
ELSE
    SEND (number & " is not a prime number") TO DISPLAY
END IF
```

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15. Question 2 · part a.i · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a.i only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(a) Answer these questions about the pseudocode.

(i) Identify a line number where **selection** starts.

(1)

(ii) Identify the line number where **iteration** starts.

(1)

(iii) Identify a line number where a **string** and a **number** are printed on the same line.

(1)

(iv) Identify the name of a **local variable**.

(1)

(v) Identify the name of a **parameter**.

(1)

(b) Write a program to implement the logic in the pseudocode.

Open **Q02b** in the code editor.

Write the program.

You must use the structure given in **Q02b** to complete the program.

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(11)

(Total for Question 2 = 16 marks)

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15. Question 2 · part a.i · 1 mark · Working and answer

Working and answer

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16. Question 2 · part a.ii · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a.ii only.

2 Raza is writing a program to tell the user whether a number they input is a prime number.

A prime number is a whole number, larger than one, that can only be divided by one and itself with no remainder.

This pseudocode contains the logic required to complete the program.

```
FUNCTION checkPrime(pNumber)
BEGIN FUNCTION
    IF pNumber = 1 THEN
        check = False
    ELSE
        check = True
        FOR count FROM 2 TO pNumber DO
            IF pNumber MOD count = 0 THEN
                check = False
            END IF
        END FOR
    END IF
RETURN check
END FUNCTION

SEND "Enter a number: " TO DISPLAY
RECEIVE number FROM (INTEGER) KEYBOARD
SET result TO checkPrime(number)

IF result = True THEN
    SEND (number & " is a prime number") TO DISPLAY
ELSE
    SEND (number & " is not a prime number") TO DISPLAY
END IF
```

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DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 7 2 4 0 6 A 0 4 1 6

16. Question 2 · part a.ii · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(a) Answer these questions about the pseudocode.

(i) Identify a line number where **selection** starts.

(1)

(ii) Identify the line number where **iteration** starts.

(1)

(iii) Identify a line number where a **string** and a **number** are printed on the same line.

(1)

(iv) Identify the name of a **local variable**.

(1)

(v) Identify the name of a **parameter**.

(1)

(b) Write a program to implement the logic in the pseudocode.

Open **Q02b** in the code editor.

Write the program.

You must use the structure given in **Q02b** to complete the program.

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(11)

(Total for Question 2 = 16 marks)

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16. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

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17. Question 2 · part a.iii · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a.iii only.

2 Raza is writing a program to tell the user whether a number they input is a prime number.

A prime number is a whole number, larger than one, that can only be divided by one and itself with no remainder.

This pseudocode contains the logic required to complete the program.

```
FUNCTION checkPrime(pNumber)
BEGIN FUNCTION
    IF pNumber = 1 THEN
        check = False
    ELSE
        check = True
        FOR count FROM 2 TO pNumber DO
            IF pNumber MOD count = 0 THEN
                check = False
            END IF
        END FOR
    END IF
RETURN check
END FUNCTION

SEND "Enter a number: " TO DISPLAY
RECEIVE number FROM (INTEGER) KEYBOARD
SET result TO checkPrime(number)

IF result = True THEN
    SEND (number & " is a prime number") TO DISPLAY
ELSE
    SEND (number & " is not a prime number") TO DISPLAY
END IF
```

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17. Question 2 · part a.iii · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a.iii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(a) Answer these questions about the pseudocode.

(i) Identify a line number where **selection** starts.

(1)

(ii) Identify the line number where **iteration** starts.

(1)

(iii) Identify a line number where a **string** and a **number** are printed on the same line.

(1)

(iv) Identify the name of a **local variable**.

(1)

(v) Identify the name of a **parameter**.

(1)

(b) Write a program to implement the logic in the pseudocode.

Open **Q02b** in the code editor.

Write the program.

You must use the structure given in **Q02b** to complete the program.

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(11)

(Total for Question 2 = 16 marks)

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17. Question 2 · part a.iii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

18. Question 1 · part b.i · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part b.i only.

1 Programs are used to handle financial transactions.

(a) Monthly account statements are created by a program.

Figure 1 shows a statement.

Date	Description	Debit (£)	Credit (£)	Balance (£)
01 Jan	Opening balance			128.35
04 Jan	Music store	10.23		118.12
07 Jan	Salary		1,515.28	1,633.40
11 Jan	Electricity company	50.00		1,583.40
19 Jan	Uniform shop	12.56		1,570.84
27 Jan	Refund		25.00	1,595.84
29 Jan	Rent	750.00		845.84
31 Jan	Interest		0.25	846.09

Figure 1

Complete the table to identify an input, an output and a process used by the program to generate the **furthest right column**.

(3)

Input	
Processing	
Output	

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DO NOT WRITE IN THIS AREA

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18. Question 1 · part b.i · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part b.i only.

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DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) A tax rate is applied to a gross value to give a net value.

Open **Q01b** in the code editor.

Use the code to answer these questions.

(i) Give the contents of a comment.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Give the name of a global variable.

(1)

(iv) Give the name of a constant.

(1)

(v) Give the name of a parameter.

(1)

(c) Variables and constants are used in program code.

(i) State the purpose of a constant.

(1)

(ii) Give **one** benefit of using a constant.

(1)

(Total for Question 1 = 10 marks)

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18. Question 1 · part b.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

19. Question 1 · part b.ii · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part b.ii only.

1 Programs are used to handle financial transactions.

(a) Monthly account statements are created by a program.

Figure 1 shows a statement.

Date	Description	Debit (£)	Credit (£)	Balance (£)
01 Jan	Opening balance			128.35
04 Jan	Music store	10.23		118.12
07 Jan	Salary		1,515.28	1,633.40
11 Jan	Electricity company	50.00		1,583.40
19 Jan	Uniform shop	12.56		1,570.84
27 Jan	Refund		25.00	1,595.84
29 Jan	Rent	750.00		845.84
31 Jan	Interest		0.25	846.09

Figure 1

Complete the table to identify an input, an output and a process used by the program to generate the **furthest right column**.

(3)

Input	
Processing	
Output	

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OLevels.com SCHOOL

19. Question 1 · part b.ii · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) A tax rate is applied to a gross value to give a net value.

Open **Q01b** in the code editor.

Use the code to answer these questions.

(i) Give the contents of a comment.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Give the name of a global variable.

(1)

(iv) Give the name of a constant.

(1)

(v) Give the name of a parameter.

(1)

(c) Variables and constants are used in program code.

(i) State the purpose of a constant.

(1)

(ii) Give **one** benefit of using a constant.

(1)

(Total for Question 1 = 10 marks)

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19. Question 1 · part b.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

20. Question 1 · part c.i · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part c.i only.

1 Programmers write programs to solve problems.

(a) Programmers use many different techniques and tools to solve problems.

Identify what is meant by the term **abstraction**.

(1)

- ☒ **A** Breaking a problem down into smaller more manageable parts
- ☒ **B** Deciding if a program is efficient in terms of execution time
- ☒ **C** Drawing flowcharts with symbols used in the computer industry
- ☒ **D** Removing unnecessary detail to highlight the important points

(b) Open **Q01b** in the code editor.

A user enters a name and an age when the program executes.

The program should display a welcome message when the user is less than 30 years of age.

Amend the code to complete the program.

Save your amended code as **Q01bFINISHED** with the correct file extension for the programming language.

(4)

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20. Question 1 · part c.i · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part c.i only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

(c) Open **Q01c** in the code editor.

Use the code to answer these questions.

(i) Give the text of a comment used in the code.

(1)

(ii) Give the text of a line that creates and initialises a variable.

(1)

(iii) Give the keyword that starts the selection used in the code.

(1)

(iv) Give the logical operator used in the code.

(1)

(d) Arrays and records are data structures.

Identify the data that **must be** stored in a record rather than an array.

(1)

- ☒ **A** "King Lear", "Macbeth", "Romeo and Juliet", "The Tempest"
- ☒ **B** "King Lear", 1983, 78.32, 99
- ☒ **C** 1978, 1985, 1990, 2001
- ☒ **D** 13.99, 12.75, 10.58, 11.43

(e) Complete the table to show the data type for **each** item.

(2)

Item	Data type
45.82	
True	

(Total for Question 1 = 12 marks)

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20. Question 1 · part c.i · 1 mark · Working and answer

Working and answer

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21. Question 1 · part c.ii · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part c.ii only.

1 Programmers write programs to solve problems.

(a) Programmers use many different techniques and tools to solve problems.

Identify what is meant by the term **abstraction**.

(1)

- ☒ **A** Breaking a problem down into smaller more manageable parts
- ☒ **B** Deciding if a program is efficient in terms of execution time
- ☒ **C** Drawing flowcharts with symbols used in the computer industry
- ☒ **D** Removing unnecessary detail to highlight the important points

(b) Open **Q01b** in the code editor.

A user enters a name and an age when the program executes.

The program should display a welcome message when the user is less than 30 years of age.

Amend the code to complete the program.

Save your amended code as **Q01bFINISHED** with the correct file extension for the programming language.

(4)

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DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

21. Question 1 · part c.ii · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part c.ii only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

(c) Open **Q01c** in the code editor.

Use the code to answer these questions.

(i) Give the text of a comment used in the code.

(1)

(ii) Give the text of a line that creates and initialises a variable.

(1)

(iii) Give the keyword that starts the selection used in the code.

(1)

(iv) Give the logical operator used in the code.

(1)

(d) Arrays and records are data structures.

Identify the data that **must be** stored in a record rather than an array.

(1)

- ☒ **A** "King Lear", "Macbeth", "Romeo and Juliet", "The Tempest"
- ☒ **B** "King Lear", 1983, 78.32, 99
- ☒ **C** 1978, 1985, 1990, 2001
- ☒ **D** 13.99, 12.75, 10.58, 11.43

(e) Complete the table to show the data type for **each** item.

(2)

Item	Data type
45.82	
True	

(Total for Question 1 = 12 marks)

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21. Question 1 · part c.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

22. Question 1 · part c.iii · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part c.iii only.

1 Programmers write programs to solve problems.

(a) Programmers use many different techniques and tools to solve problems.

Identify what is meant by the term **abstraction**.

(1)

- ☒ **A** Breaking a problem down into smaller more manageable parts
- ☒ **B** Deciding if a program is efficient in terms of execution time
- ☒ **C** Drawing flowcharts with symbols used in the computer industry
- ☒ **D** Removing unnecessary detail to highlight the important points

(b) Open **Q01b** in the code editor.

A user enters a name and an age when the program executes.

The program should display a welcome message when the user is less than 30 years of age.

Amend the code to complete the program.

Save your amended code as **Q01bFINISHED** with the correct file extension for the programming language.

(4)

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DO NOT WRITE IN THIS AREA

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22. Question 1 · part c.iii · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part c.iii only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

(c) Open **Q01c** in the code editor.

Use the code to answer these questions.

(i) Give the text of a comment used in the code.

(1)

(ii) Give the text of a line that creates and initialises a variable.

(1)

(iii) Give the keyword that starts the selection used in the code.

(1)

(iv) Give the logical operator used in the code.

(1)

(d) Arrays and records are data structures.

Identify the data that **must be** stored in a record rather than an array.

(1)

- ☒ **A** "King Lear", "Macbeth", "Romeo and Juliet", "The Tempest"
- ☒ **B** "King Lear", 1983, 78.32, 99
- ☒ **C** 1978, 1985, 1990, 2001
- ☒ **D** 13.99, 12.75, 10.58, 11.43

(e) Complete the table to show the data type for **each** item.

(2)

Item	Data type
45.82	
True	

(Total for Question 1 = 12 marks)

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22. Question 1 · part c.iii · 1 mark · Working and answer

Working and answer

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23. Question 1 · part d.ii · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part d.ii only.

1 Programmers use different programming constructs to create working code.

(a) Identify the keyword used to create a selection statement.

(1)

- ☒ **A** for
- ☒ **B** if
- ☒ **C** new
- ☒ **D** while

(b) Computer code can contain errors.

(i) Complete the table by adding a tick (✓) to match each error description to a type of error.

(3)

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.			
The program does not translate.			
The program crashes during execution.			

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DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

23. Question 1 · part d.ii · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part d.ii only.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

(ii) A program counts the number of green and red balls stored in an array.

The program outputs the number of green balls, the number of red balls and the total number of balls.

Figure 1 shows the expected output from the program.

```
Green:  25 balls
Red:  15 balls
Total:  40 balls
```

Figure 1

Open **Q01bii** in the code editor.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01biiFINISHED** with the correct file extension for the programming language.

(3)

(c) Programs use constants, variables and data structures.

(i) Give **one** reason for using a constant.

(1)

(ii) Describe **one** difference between a variable and an array.

(2)

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23. Question 1 · part d.ii · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part d.ii only.

(d) A program is used to calculate wages for sales staff. Staff can earn 5% commission based on their age and sales.

Open **Q01d** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a user-written subprogram.

(1)

.....

(ii) Identify a logical operator used in this program.

(1)

.....

(iii) Identify the name of a variable that holds a real number.

(1)

.....

(iv) Identify the name of a parameter.

(1)

.....

(Total for Question 1 = 14 marks)

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Working and answer

Practice workbook · Find named items in a program

Reading an opened or printed program and naming an identifier by its role: numeric/string/real variable, local or global variable, constant, data structure, user-written or built-in subprogram, parameter, argument passed, variable receiving a return value.

18 question parts: 1 worked-example retry and 17 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T2-12

1. Question 1 · part d.iv · 1 mark

2025 JUN · Paper 02

Worked-example retry · Attempt part d.iv only.

1 Programmers use different programming constructs to create working code.

(a) Identify the keyword used to create a selection statement.

(1)

- ☒ **A** for
- ☒ **B** if
- ☒ **C** new
- ☒ **D** while

(b) Computer code can contain errors.

(i) Complete the table by adding a tick (✓) to match each error description to a type of error.

(3)

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.			
The program does not translate.			
The program crashes during execution.			

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DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

1. Question 1 · part d.iv · 1 mark

2025 JUN · Paper 02

Worked-example retry · Attempt part d.iv only.

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DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(ii) A program counts the number of green and red balls stored in an array.

The program outputs the number of green balls, the number of red balls and the total number of balls.

Figure 1 shows the expected output from the program.

```
Green:  25 balls
Red:  15 balls
Total:  40 balls
```

Figure 1

Open **Q01bii** in the code editor.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01biiFINISHED** with the correct file extension for the programming language.

(3)

(c) Programs use constants, variables and data structures.

(i) Give **one** reason for using a constant.

(1)

(ii) Describe **one** difference between a variable and an array.

(2)

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1. Question 1 · part d.iv · 1 mark

2025 JUN · Paper 02

Worked-example retry · Attempt part d.iv only.

(d) A program is used to calculate wages for sales staff. Staff can earn 5% commission based on their age and sales.

Open **Q01d** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a user-written subprogram.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Identify the name of a variable that holds a real number.

(1)

(iv) Identify the name of a parameter.

(1)

(Total for Question 1 = 14 marks)

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Working and answer

2. Question 2 · part c.iii · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.iii only.

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2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

2. Question 2 · part c.iii · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.iii only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

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2. Question 2 · part c.iii · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.iii only.

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(c) Anna has written a program that manipulates text.

Open **Q02c** in the code editor.

Use the code to answer these questions.

(i) Identify a line number where the code includes a relational operator.

(1)

(ii) Identify a line number where a variable is set to a numeric value entered by a user.

(1)

(iii) Identify the name of a variable passed from the main program to the subprogram.

(1)

(iv) Identify the name of a variable that receives the value calculated by the subprogram.

(1)

(v) State the value returned by the subprogram when the input values are **educational**, **3**, **6**

(2)

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2. Question 2 · part c.iii · 1 mark · Working and answer

Working and answer

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3. Question 2 · part c.iv · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.iv only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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3. Question 2 · part c.iv · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.iv only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

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3. Question 2 · part c.iv · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.iv only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

(c) Anna has written a program that manipulates text.

Open **Q02c** in the code editor.

Use the code to answer these questions.

(i) Identify a line number where the code includes a relational operator.

(1)

(ii) Identify a line number where a variable is set to a numeric value entered by a user.

(1)

(iii) Identify the name of a variable passed from the main program to the subprogram.

(1)

(iv) Identify the name of a variable that receives the value calculated by the subprogram.

(1)

(v) State the value returned by the subprogram when the input values are **educational**, **3**, **6**

(2)

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3. Question 2 · part c.iv · 1 mark · Working and answer

Working and answer

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4. Question 1 · part e.iii · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part e.iii only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is **not** an arithmetic operator.

(1)

- ☒ **A** add
- ☒ **B** divide
- ☒ **C** equals
- ☒ **D** modulus

(b) State the values used by the Boolean data type.

(1)

.....

.....

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4. Question 1 · part e.iii · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part e.iii only.

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DO NOT WRITE IN THIS AREA

(c) Open **Q01c** in the code editor.

The program should multiply two integers.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) State what is meant by the term 'variable'.

(1)

(e) Ayesha has written a program that calculates the sum of a set of numbers.

Open **Q01e** in the code editor.

Answer these questions about the code.

(i) Identify the line number where the code includes a logic operator.

(1)

(ii) Identify the line number where a subprogram definition starts.

(1)

(iii) Identify the name of a numeric variable.

(1)

(iv) Identify the name of a string variable.

(1)

(Total for Question 1 = 10 marks)

4. Question 1 · part e.iii · 1 mark · Working and answer

Working and answer

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5. Question 1 · part e.iv · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part e.iv only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is **not** an arithmetic operator.

(1)

- ☒ **A** add
- ☒ **B** divide
- ☒ **C** equals
- ☒ **D** modulus

(b) State the values used by the Boolean data type.

(1)

.....

.....

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DO NOT WRITE IN THIS AREA

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5. Question 1 · part e.iv · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part e.iv only.

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DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(c) Open **Q01c** in the code editor.

The program should multiply two integers.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) State what is meant by the term 'variable'.

(1)

(e) Ayesha has written a program that calculates the sum of a set of numbers.

Open **Q01e** in the code editor.

Answer these questions about the code.

(i) Identify the line number where the code includes a logic operator.

(1)

(ii) Identify the line number where a subprogram definition starts.

(1)

(iii) Identify the name of a numeric variable.

(1)

(iv) Identify the name of a string variable.

(1)

(Total for Question 1 = 10 marks)

5. Question 1 · part e.iv · 1 mark · Working and answer

Working and answer

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6. Question 3 · part d.ii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.ii only.

3 Several encryption algorithms have been developed.

(a) Ahmed has started to create a Caesar cipher encryption program.

Caesar cipher encryption works by giving a number value to a key.

Each letter in a plaintext message is replaced by a new ciphertext letter using the key.

For example, as shown in the table, a key value of +2 would change the plaintext message **encrypt** to the ciphertext message **gpetarv**.

Plaintext	e	n	c	r	y	p	t
Ciphertext	g	p	e	t	a	r	v

Ahmed's program already converts from plaintext to ciphertext.

He wants to extend the program to:

- accept the input of a plaintext message
- accept the input of the key, which must be a positive integer between 1 and 25
- write the generated ciphertext to a text file named **Cipher.txt**
- display the ciphertext.

Open **Q03a** in the code editor.

Complete the program code.

You must use the structure given in the file **Q03a** to complete the program.

Do not add any further functionality.

Save your code as **Q03aFINISHED** with the correct file extension for the programming language.

(8)

(b) Explain **one** reason why the Caesar cipher is a weak encryption algorithm.

(2)

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6. Question 3 · part d.ii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.ii only.

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(c) The Vigenere cipher encrypts plaintext using a series of Caesar ciphers, based on the letters of a keyword and a Vigenere square.

Figure 1 shows a Vigenere square.

	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
b	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a
c	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b
d	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c
e	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d
f	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e
g	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f
h	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g
i	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h
j	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i
k	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j
l	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k
m	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l
n	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m
o	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n
p	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
q	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p
r	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q
s	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r
t	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s
u	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t
v	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u
w	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v
x	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w
y	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x
z	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y

Figure 1

Complete the table to perform a Vigenere decryption of the ciphertext message **flmktrwhee** using the keyword **thirty**.

(4)

Encrypted letter	f	l	m	k	t	r	w	h	e	e
Keyword letter										
Decrypted letter										

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6. Question 3 · part d.ii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.ii only.

(d) A user-defined function uses a Vigenere cipher to encrypt plaintext into ciphertext.

This pseudocode shows how the function is called.

```
SET ciphertext TO ciphertext & getCipherLetter(keywordLetter, plaintextLetter)
```

(i) State what would be returned by this function.

(1)

.....

.....

(ii) Give the name of **one** parameter passed to this function.

(1)

.....

(iii) State what is meant by a **built-in subprogram**.

(1)

.....

.....

(Total for Question 3 = 17 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE I

O1 levels.com SCHOOL

Working and answer

7. Question 3 · part a.i · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part a.i only.

3 Dates are often used in programs.

(a) A program accepts the number of a month and outputs the number of days in the month.

Open **Q03a** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a local variable.

(1)

(ii) Identify the name of a data structure.

(1)

(iii) Identify the name of a built-in subprogram.

(1)

(iv) Identify the name of a user-written function.

(1)

(v) Identify the name of a parameter.

(1)

(vi) The program includes validation to ensure the month number is between 1 and 12 inclusive.

Complete the table by adding the type of test data that is being used to test the validation.

The first type of test has been completed for you.

(2)

Type of test data	Test data
Normal	3
	1
	15

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

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7. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 3 · part a.ii · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part a.ii only.

3 Dates are often used in programs.

(a) A program accepts the number of a month and outputs the number of days in the month.

Open **Q03a** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a local variable.

(1)

(ii) Identify the name of a data structure.

(1)

(iii) Identify the name of a built-in subprogram.

(1)

(iv) Identify the name of a user-written function.

(1)

(v) Identify the name of a parameter.

(1)

(vi) The program includes validation to ensure the month number is between 1 and 12 inclusive.

Complete the table by adding the type of test data that is being used to test the validation.

The first type of test has been completed for you.

(2)

Type of test data	Test data
Normal	3
	1
	15

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

8. Question 3 · part a.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 3 · part a.iii · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part a.iii only.

3 Dates are often used in programs.

(a) A program accepts the number of a month and outputs the number of days in the month.

Open **Q03a** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a local variable.

(1)

(ii) Identify the name of a data structure.

(1)

(iii) Identify the name of a built-in subprogram.

(1)

(iv) Identify the name of a user-written function.

(1)

(v) Identify the name of a parameter.

(1)

(vi) The program includes validation to ensure the month number is between 1 and 12 inclusive.

Complete the table by adding the type of test data that is being used to test the validation.

The first type of test has been completed for you.

(2)

Type of test data	Test data
Normal	3
	1
	15

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

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9. Question 3 · part a.iii · 1 mark · Working and answer

Working and answer

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10. Question 3 · part a.iv · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part a.iv only.

3 Dates are often used in programs.

(a) A program accepts the number of a month and outputs the number of days in the month.

Open **Q03a** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a local variable.

(1)

(ii) Identify the name of a data structure.

(1)

(iii) Identify the name of a built-in subprogram.

(1)

(iv) Identify the name of a user-written function.

(1)

(v) Identify the name of a parameter.

(1)

(vi) The program includes validation to ensure the month number is between 1 and 12 inclusive.

Complete the table by adding the type of test data that is being used to test the validation.

The first type of test has been completed for you.

(2)

Type of test data	Test data
Normal	3
	1
	15

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

10. Question 3 · part a.iv · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 3 · part a.v · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part a.v only.

3 Dates are often used in programs.

(a) A program accepts the number of a month and outputs the number of days in the month.

Open **Q03a** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a local variable.

(1)

(ii) Identify the name of a data structure.

(1)

(iii) Identify the name of a built-in subprogram.

(1)

(iv) Identify the name of a user-written function.

(1)

(v) Identify the name of a parameter.

(1)

(vi) The program includes validation to ensure the month number is between 1 and 12 inclusive.

Complete the table by adding the type of test data that is being used to test the validation.

The first type of test has been completed for you.

(2)

Type of test data	Test data
Normal	3
	1
	15

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

11. Question 3 · part a.v · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 2 · part a.iv · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a.iv only.

2 Raza is writing a program to tell the user whether a number they input is a prime number.

A prime number is a whole number, larger than one, that can only be divided by one and itself with no remainder.

This pseudocode contains the logic required to complete the program.

```
FUNCTION checkPrime(pNumber)
BEGIN FUNCTION
    IF pNumber = 1 THEN
        check = False
    ELSE
        check = True
        FOR count FROM 2 TO pNumber DO
            IF pNumber MOD count = 0 THEN
                check = False
            END IF
        END FOR
    END IF
RETURN check
END FUNCTION

SEND "Enter a number: " TO DISPLAY
RECEIVE number FROM (INTEGER) KEYBOARD
SET result TO checkPrime(number)

IF result = True THEN
    SEND (number & " is a prime number") TO DISPLAY
ELSE
    SEND (number & " is not a prime number") TO DISPLAY
END IF
```

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

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12. Question 2 · part a.iv · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a.iv only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(a) Answer these questions about the pseudocode.

(i) Identify a line number where **selection** starts.

(1)

(ii) Identify the line number where **iteration** starts.

(1)

(iii) Identify a line number where a **string** and a **number** are printed on the same line.

(1)

(iv) Identify the name of a **local variable**.

(1)

(v) Identify the name of a **parameter**.

(1)

(b) Write a program to implement the logic in the pseudocode.

Open **Q02b** in the code editor.

Write the program.

You must use the structure given in **Q02b** to complete the program.

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(11)

(Total for Question 2 = 16 marks)

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12. Question 2 · part a.iv · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

13. Question 2 · part a.v · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a.v only.

2 Raza is writing a program to tell the user whether a number they input is a prime number.

A prime number is a whole number, larger than one, that can only be divided by one and itself with no remainder.

This pseudocode contains the logic required to complete the program.

```
FUNCTION checkPrime(pNumber)
BEGIN FUNCTION
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        check = False
    ELSE
        check = True
        FOR count FROM 2 TO pNumber DO
            IF pNumber MOD count = 0 THEN
                check = False
            END IF
        END FOR
    END IF
RETURN check
END FUNCTION

SEND "Enter a number: " TO DISPLAY
RECEIVE number FROM (INTEGER) KEYBOARD
SET result TO checkPrime(number)

IF result = True THEN
    SEND (number & " is a prime number") TO DISPLAY
ELSE
    SEND (number & " is not a prime number") TO DISPLAY
END IF
```

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

13. Question 2 · part a.v · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a.v only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(a) Answer these questions about the pseudocode.

(i) Identify a line number where **selection** starts.

(1)

(ii) Identify the line number where **iteration** starts.

(1)

(iii) Identify a line number where a **string** and a **number** are printed on the same line.

(1)

(iv) Identify the name of a **local variable**.

(1)

(v) Identify the name of a **parameter**.

(1)

(b) Write a program to implement the logic in the pseudocode.

Open **Q02b** in the code editor.

Write the program.

You must use the structure given in **Q02b** to complete the program.

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(11)

(Total for Question 2 = 16 marks)

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13. Question 2 · part a.v · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

14. Question 1 · part b.iii · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part b.iii only.

1 Programs are used to handle financial transactions.

(a) Monthly account statements are created by a program.

Figure 1 shows a statement.

Date	Description	Debit (£)	Credit (£)	Balance (£)
01 Jan	Opening balance			128.35
04 Jan	Music store	10.23		118.12
07 Jan	Salary		1,515.28	1,633.40
11 Jan	Electricity company	50.00		1,583.40
19 Jan	Uniform shop	12.56		1,570.84
27 Jan	Refund		25.00	1,595.84
29 Jan	Rent	750.00		845.84
31 Jan	Interest		0.25	846.09

Figure 1

Complete the table to identify an input, an output and a process used by the program to generate the **furthest right column**.

(3)

Input	
Processing	
Output	

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

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14. Question 1 · part b.iii · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part b.iii only.

DO NOT WRITE IN THIS AREA

(b) A tax rate is applied to a gross value to give a net value.

Open **Q01b** in the code editor.

Use the code to answer these questions.

(i) Give the contents of a comment.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Give the name of a global variable.

(1)

(iv) Give the name of a constant.

(1)

(v) Give the name of a parameter.

(1)

(c) Variables and constants are used in program code.

(i) State the purpose of a constant.

(1)

(ii) Give **one** benefit of using a constant.

(1)

(Total for Question 1 = 10 marks)

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14. Question 1 · part b.iii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

15. Question 1 · part b.iv · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part b.iv only.

1 Programs are used to handle financial transactions.

(a) Monthly account statements are created by a program.

Figure 1 shows a statement.

Date	Description	Debit (£)	Credit (£)	Balance (£)
01 Jan	Opening balance			128.35
04 Jan	Music store	10.23		118.12
07 Jan	Salary		1,515.28	1,633.40
11 Jan	Electricity company	50.00		1,583.40
19 Jan	Uniform shop	12.56		1,570.84
27 Jan	Refund		25.00	1,595.84
29 Jan	Rent	750.00		845.84
31 Jan	Interest		0.25	846.09

Figure 1

Complete the table to identify an input, an output and a process used by the program to generate the **furthest right column**.

(3)

Input	
Processing	
Output	

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

15. Question 1 · part b.iv · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part b.iv only.

DO NOT WRITE IN THIS AREA

(b) A tax rate is applied to a gross value to give a net value.

Open **Q01b** in the code editor.

Use the code to answer these questions.

(i) Give the contents of a comment.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Give the name of a global variable.

(1)

(iv) Give the name of a constant.

(1)

(v) Give the name of a parameter.

(1)

(c) Variables and constants are used in program code.

(i) State the purpose of a constant.

(1)

(ii) Give **one** benefit of using a constant.

(1)

(Total for Question 1 = 10 marks)

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15. Question 1 · part b.iv · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

16. Question 1 · part b.v · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part b.v only.

1 Programs are used to handle financial transactions.

(a) Monthly account statements are created by a program.

Figure 1 shows a statement.

Date	Description	Debit (£)	Credit (£)	Balance (£)
01 Jan	Opening balance			128.35
04 Jan	Music store	10.23		118.12
07 Jan	Salary		1,515.28	1,633.40
11 Jan	Electricity company	50.00		1,583.40
19 Jan	Uniform shop	12.56		1,570.84
27 Jan	Refund		25.00	1,595.84
29 Jan	Rent	750.00		845.84
31 Jan	Interest		0.25	846.09

Figure 1

Complete the table to identify an input, an output and a process used by the program to generate the **furthest right column**.

(3)

Input	
Processing	
Output	

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

16. Question 1 · part b.v · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part b.v only.

DO NOT WRITE IN THIS AREA

(b) A tax rate is applied to a gross value to give a net value.

Open **Q01b** in the code editor.

Use the code to answer these questions.

(i) Give the contents of a comment.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Give the name of a global variable.

(1)

(iv) Give the name of a constant.

(1)

(v) Give the name of a parameter.

(1)

(c) Variables and constants are used in program code.

(i) State the purpose of a constant.

(1)

(ii) Give **one** benefit of using a constant.

(1)

(Total for Question 1 = 10 marks)

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16. Question 1 · part b.v · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

17. Question 1 · part d.i · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part d.i only.

1 Programmers use different programming constructs to create working code.

(a) Identify the keyword used to create a selection statement.

(1)

- ☒ **A** for
- ☒ **B** if
- ☒ **C** new
- ☒ **D** while

(b) Computer code can contain errors.

(i) Complete the table by adding a tick (✓) to match each error description to a type of error.

(3)

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.			
The program does not translate.			
The program crashes during execution.			

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DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

17. Question 1 · part d.i · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part d.i only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(ii) A program counts the number of green and red balls stored in an array.

The program outputs the number of green balls, the number of red balls and the total number of balls.

Figure 1 shows the expected output from the program.

```
Green:  25 balls
Red:  15 balls
Total:  40 balls
```

Figure 1

Open **Q01bii** in the code editor.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01biiFINISHED** with the correct file extension for the programming language.

(3)

(c) Programs use constants, variables and data structures.

(i) Give **one** reason for using a constant.

(1)

(ii) Describe **one** difference between a variable and an array.

(2)

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17. Question 1 · part d.i · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part d.i only.

(d) A program is used to calculate wages for sales staff. Staff can earn 5% commission based on their age and sales.

Open **Q01d** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a user-written subprogram.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Identify the name of a variable that holds a real number.

(1)

(iv) Identify the name of a parameter.

(1)

(Total for Question 1 = 14 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN

OLevels.COM SCHOOL

Working and answer

18. Question 1 · part d.iii · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part d.iii only.

1 Programmers use different programming constructs to create working code.

(a) Identify the keyword used to create a selection statement.

(1)

- ☒ **A** for
- ☒ **B** if
- ☒ **C** new
- ☒ **D** while

(b) Computer code can contain errors.

(i) Complete the table by adding a tick (✓) to match each error description to a type of error.

(3)

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.			
The program does not translate.			
The program crashes during execution.			

OLevels.COM SCHOOL

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

18. Question 1 · part d.iii · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part d.iii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(ii) A program counts the number of green and red balls stored in an array.

The program outputs the number of green balls, the number of red balls and the total number of balls.

Figure 1 shows the expected output from the program.

```
Green:  25 balls
Red:  15 balls
Total:  40 balls
```

Figure 1

Open **Q01bii** in the code editor.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01biiFINISHED** with the correct file extension for the programming language.

(3)

(c) Programs use constants, variables and data structures.

(i) Give **one** reason for using a constant.

(1)

(ii) Describe **one** difference between a variable and an array.

(2)

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18. Question 1 · part d.iii · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part d.iii only.

(d) A program is used to calculate wages for sales staff. Staff can earn 5% commission based on their age and sales.

Open **Q01d** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a user-written subprogram.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Identify the name of a variable that holds a real number.

(1)

(iv) Identify the name of a parameter.

(1)

(Total for Question 1 = 14 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN

OLevels.COM SCHOOL

Working and answer

Practice workbook · Normal, boundary and erroneous test data

Choosing, labelling and contrasting normal, boundary and erroneous test data for a stated rule or program.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T2-09

1. Question 2 · part c.i · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part c.i only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

2 Programs can process numerical data.

(a) Identify the method used to check input data matches set rules.

(1)

- ☒ **A** Abstraction
- ☒ **B** Encryption
- ☒ **C** Iteration
- ☒ **D** Validation

(b) **Figure 2** shows a section from an employee payslip produced by a computer program.

Pay Rate	Hours		Total Pay
£ 18.75	19		£ 356.25
	Tax	20%	£ 71.25
		Income	£ 285.00

Figure 2

(i) Complete the table to give **one** data type shown in **Figure 2** and give an example value.

(2)

Data type	
Example	

(ii) Complete the table to give an input, a process and an output used in the program to produce the payslip.

(3)

Input	
Process	
Output	

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1. Question 2 · part c.i · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part c.i only.

(c) Numerical values can be represented in denary, binary and hexadecimal.

A program converts denary values in the range 0 to 255 into hexadecimal.

(i) The program is tested using appropriate integer values.

Complete the table to show an example of normal, boundary and erroneous test data for this program.

(2)

	Test data
Normal	
Boundary	
Erroneous	

(ii) The program uses subprograms to do the conversion.

Open **Q02cii** in the code editor.

Amend the code to complete the program to:

- convert the user input to integer
- check the input is between 0 and 255
- call the **getHex** subprogram, with the user input as an argument.

Do not add any further functionality.

Save your code as **Q02ciiFINISHED** with the correct file extension for the programming language.

(4)

(Total for Question 2 = 12 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

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1. Question 2 · part c.i · 2 marks · Working and answer

Working and answer

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2. Question 2 · part c · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part c only.

2 A book club uses computer applications.

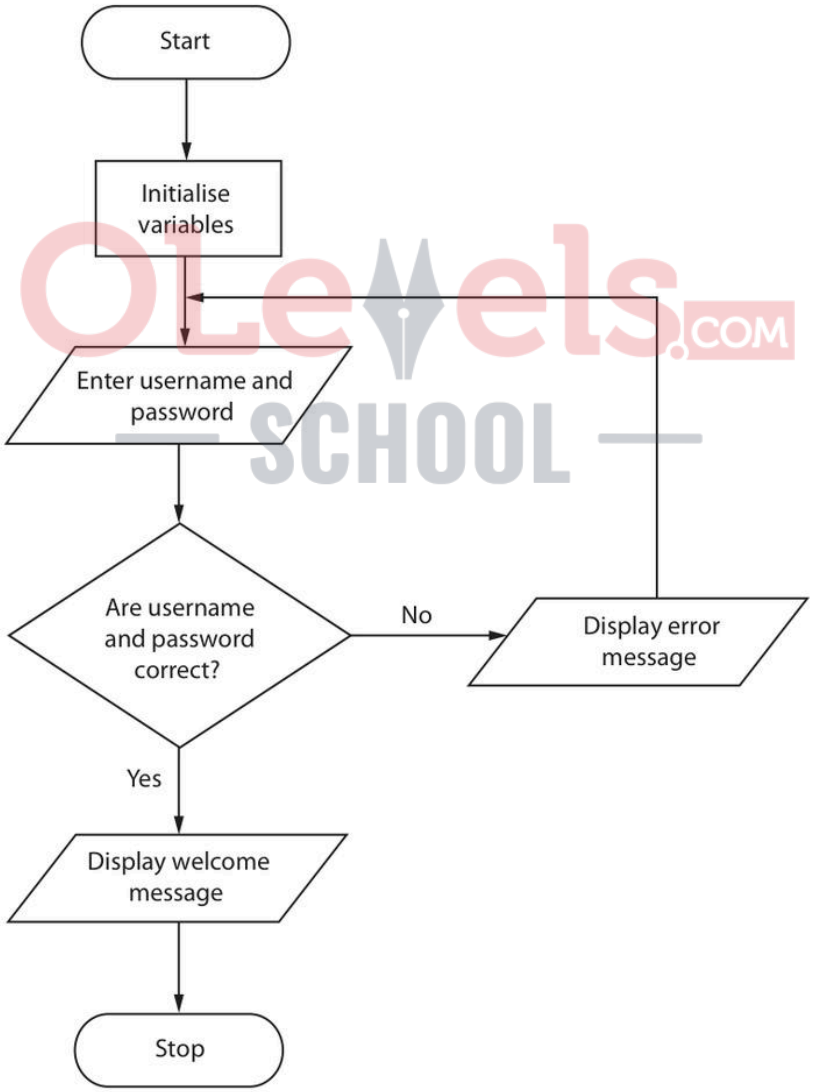
(a) The club wants a program to ensure that logins are valid.

A flowchart for a test version of the program has been designed.

The test version uses:

- a username of bard423
- a password of nX2934?

Here is the flowchart.



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2. Question 2 · part c · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part c only.

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Write a program to implement the logic in the flowchart.

Open **Q02a** in the code editor.

You must use the structure given in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(11)

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2. Question 2 · part c · 3 marks

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Further practice · Attempt part c only.

(b) The club sells books.

It needs a computer program to monitor:

- the number of books sold
- the amount of profit made.

Q02b provides a structure for the program.

Open **Q02b** in the code editor.

Amend the code to complete the If statement used to produce the outputs described in the table.

Condition	Output message
Number of books sold is at least 5 and profit made is at least 10	Sales and profit are good this week
Number of books sold is over 20 and profit made is at least 20	Sales and profit are excellent this week
Number of books sold is under 5 or profit made is under 5	Poor performance this week
All other inputs	Alert manager

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(4)

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2. Question 2 · part c · 3 marks

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Further practice · Attempt part c only.

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(c) Boundary testing is an important part of program development.

The boundary value is the most extreme value that will pass the test.

For example, if the boundary condition is no more than 30 then the boundary test data would be 30.

Complete the test table to show the boundary test data that could be used for each test.

(3)

	Test data	Expected output
booksSold		Poor performances this week
profit		
booksSold		Sales and profit are good this week
profit		
booksSold		Sales and profit are excellent this week
profit		

(Total for Question 2 = 18 marks)

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Working and answer

3. Question 3 · part a.vi · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part a.vi only.

3 Dates are often used in programs.

(a) A program accepts the number of a month and outputs the number of days in the month.

Open **Q03a** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a local variable.

(1)

(ii) Identify the name of a data structure.

(1)

(iii) Identify the name of a built-in subprogram.

(1)

(iv) Identify the name of a user-written function.

(1)

(v) Identify the name of a parameter.

(1)

(vi) The program includes validation to ensure the month number is between 1 and 12 inclusive.

Complete the table by adding the type of test data that is being used to test the validation.

The first type of test has been completed for you.

(2)

Type of test data	Test data
Normal	3
	1
	15

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3. Question 3 · part a.vi · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part a.vi only.

(b) Explain **one** reason why parameters are used to pass data into subprograms.

(2)

(c) A leap year is a year that has 366 days rather than 365 days. A leap year usually occurs every four years.

A year that is a leap year:

- can be exactly divided by 400

Or

- can be exactly divided by 4, but **not** exactly divided by 100.

These years are leap years: 2000, 2012 and 2020.

These years are not leap years: 1700, 1900 and 2021.

A program is needed that will accept the input of a year and output whether it is a leap year or not.

Open **Q03c** in the code editor.

Write the program.

You **must** use the structure given in **Q03c**.

Do not add any further functionality.

Save your code as **Q03cFINISHED** with the correct file extension for the programming language.

(6)

(Total for Question 3 = 15 marks)

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3. Question 3 · part a.vi · 2 marks · Working and answer

Working and answer

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4. Question 1 · part d · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part d only.

1 Programmers write code to solve problems.

(a) Identify the description of a variable in a computer program.

(1)

- ☒ **A** A value that cannot be used more than once
- ☒ **B** A value that must be input
- ☒ **C** A value that is always used in a calculation
- ☒ **D** A value that can change

(b) Identify the technique that improves the readability of code.

(1)

- ☒ **A** Using indents on every line
- ☒ **B** Using descriptive names for variables
- ☒ **C** Using the correct operators
- ☒ **D** Using suitable data structures

(c) Complete the table by giving an example value for **each** data type.

The first row has been completed for you.

(2)

Data type	Example value
integer	12
char	
real	

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4. Question 1 · part d · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part d only.

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(d) Describe **one** difference between the data used in boundary testing and the data used in erroneous testing.

(2)

(e) A program should output the value 2

However, there is an error in the code and the actual output is 4

Name this type of error.

(1)

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(f) Open **Q01f** in the code editor.

The program should allow the input of the length of the side of a square and output the area of the square.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01fFINISHED** with the correct file extension for the programming language.

(3)

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(g) A program is needed that will accept the input of a letter and compare it with a stored letter.

It will check whether the letter comes earlier in the alphabet, later in the alphabet or is the same letter as the stored letter.

It will output the letter and the stored letter with an appropriate message.

Open **Q01g** in the code editor.

Amend the code to complete the if statement used to produce the output.

You must use the structure given in **Q01g** to complete the program.

Do not add any further functionality.

Save your code as **Q01gFINISHED** with the correct file extension for the programming language.

(4)

(Total for Question 1 = 14 marks)

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4. Question 1 · part d · 2 marks · Working and answer

Working and answer

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5. Question 2 · part b.i · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part b.i only.

2 Solutions to problems are made up of many different components.

(a) Programmers use subprograms when developing code.

(i) Give **two** reasons to use **subprogram libraries** when developing code.

(2)

1 .....

.....

2 .....

.....

(ii) Identify which **one** of these must return a result.

(1)

- ☒ **A** Function
- ☒ **B** Iteration
- ☒ **C** Procedure
- ☒ **D** Selection

(iii) State what is meant by the term **local variable**.

(1)

.....

.....

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5. Question 2 · part b.i · 3 marks

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Further practice · Attempt part b.i only.

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(b) A school is planning a fish and chip dinner for the students and their families.

(i) Tickets are only sold to families.

A family is one to four adults and one to six children.

A program is being written to help manage ticket sales.

Complete the table to show examples of normal, boundary and erroneous numeric test data for the program.

(3)

	Adult	Children
Normal		
Boundary		
Erroneous		

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(ii) The kitchen staff use a program to determine whether they have enough chips or how many more they need to order.

Open **Q02bii** in the code editor.

There are **four** errors in the code.

Amend the code to correct the errors.

Use this test data to help you find the errors.

Chips in stock (kilograms)	Number of adults	Number of children	Expected output
19	100	150	Stocks available
15	100	150	Order 4.0 kilograms

Save your amended code as **Q02biiFINISHED** with the correct file extension for the programming language.

(4)

(Total for Question 2 = 11 marks)

5. Question 2 · part b.i · 3 marks · Working and answer

Working and answer

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6. Question 2 · part d · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part d only.

2 Programmers design, correct and test programs.

(a) Programmers design programs.

(i) Identify the term for a step-by-step description to complete a task.

(1)

- ☒ **A** Algorithm
- ☒ **B** Computational thinking
- ☒ **C** Decomposition
- ☒ **D** Pseudocode

(ii) A constant is a memory location whose value does not change during program execution.

Give the name for a memory location whose value can change during program execution.

(1)

(b) Programs can have runtime errors.

(i) State what is meant by the term **runtime error**.

(1)

(ii) Give **one** example of a runtime error.

(1)

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6. Question 2 · part d · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part d only.

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(c) Open **Q02c** in the code editor.

The program generates a random number for the width of a rectangle.

The length of the rectangle is always 4

The program should calculate the perimeter of the rectangle.

The program should display the width, the length and the perimeter of the rectangle.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q02cFINISHED** with the correct file extension for the programming language.

(3)

(d) **Figure 1** shows part of an online form displayed in a browser.

Memorable word		(2-8 letters)
Date	31-01-2019	

Figure 1

Complete the table to show **two** examples of erroneous, **one** example of normal and **one** example of boundary test data for the **memorable word** field.

Each example must be different.

(4)

Erroneous	
Erroneous	
Normal	
Boundary	

(Total for Question 2 = 11 marks)

6. Question 2 · part d · 4 marks · Working and answer

Working and answer

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